EE3304 MODULE OUTLINE 2025/2026

Module Title	Electrical Circuit Analysis			
Module Code	EE3304	Semester	I	3 Credit hrs
Module Leader	Molibeli Taele			
Pre-requisites	Level 2 Physics Modules or Equivalent			
Post-requisites	All Level 3 Modules			
Course Rationale and Goals	Electrical Circuit Analysis is a foundational course in electrical and electronic engineering, providing students with the essential tools and methods for understanding and analyzing electrical systems. The course covers fundamental concepts such as voltage, current, power, and energy, and introduces systematic approaches to circuit analysis, including **Ohm's Law, Kirchhoff's Laws, the Node Voltage Method, and the Mesh Current Method**. The rationale of the course is to: • Develop the **fundamental tools of linear circuit analysis** that are essential to all engineering disciplines. • Introduce students to the **basic building blocks of electrical circuits**, including wires, resistors, capacitors, inductors, voltage and current sources, and operational amplifiers—forming the "alphabet" of circuit theory. • Prepare students for **advanced courses** in circuit analysis, electronics, and related fields by building a strong theoretical and practical foundation. The main objectives of the course are to ensure that students: 1. Understand the fundamental theory and mathematics required for the analysis of **Alternating Current (AC) circuits**. 2. Gain proficiency in analyzing **frequency response and transfer functions** of linear electrical networks. 3. Develop the ability to apply circuit analysis methods to solve **practical engineering problems**. 4. Strengthen their skills in modeling, analyzing, and interpreting the behavior of real-world electrical systems. Through this course, students will not only acquire technical knowledge but also develop the analytical mindset necessary for success in subsequent electrical engineering courses and professional practice.			
Lecture Format:	Students are expected to **review assigned reading materials prior to each class** to ensure active participation and understanding. Most class sessions will include a **short quiz** to assess comprehension of the assigned material. Class periods will focus on: • **Overview of assigned topics** • **Interactive discussions and Q&A sessions** • **Problem-solving exercises and application of concepts** This format encourages active learning, reinforces key concepts, and prepares students for both in-class and practical circuit analysis exercises			
Course Objectives	By the end of this course, students will be able to: 1. **Identify** linear electrical systems and represent them accurately in schematic form. 2. **Apply** Kirchhoff's Current Law (KCL), Kirchhoff's Voltage Law (KVL), and Ohm's Law to solve circuit problems. 3. **Simplify** circuits using series and parallel equivalents, as well as Thevenin and Norton equivalent circuits.			

	4. **Perform** node voltage and mesh current analyses, and formulate these analyses in standard matrix form. 5. **Identify and model** first- and second-order electrical systems involving capacitors and inductors. 6. **Predict** the transient response of first- and second-order circuits.

Teaching & Learning Methods	**Number/Student hours**		
Lecturers	2 hrs/week for 14 weeks	Tutorials	1 hr/week for 8 weeks
Practical	2 x 3hrs laboratory sessions		
Private Study	Reading, example problems, report writing, revision - 70 hrs.		

The course will be taught in an Engineering perspective –A different view from 2nd year physics

Method(s) and Timing of Assessment

- Several Assignments (10%): very easy for building up students' confidence, no need to copy others' work
- Midterm Test (10%), End-of-term Test (10%), Laboratory Component (20%), Final Examination (50%).

Outline Syllabus

- DC circuit analysis; units; ohm's law, Kirchhoff's Laws and their application in series and parallel circuits.
- Analysis of circuits using nodal voltages; modelling of real devices with ideal elements, equivalence of voltage and current sources.
- Thévenin's and Norton's theorem.
- Techniques of circuit analysis using mesh currents; power transfer from a source to a load.
- Linear analysis; superposition; star-delta and delta-star transformation.
- Transient effects in electrical circuits, modelling circuits by analytic solution of a differential equation.
- Exponential rise and decay, time constant in RLC circuits.
- Circuits with steady-state sinusoidal excitation; Circuit analysis suing complex mathematics; complex impedances of ideal circuit elements.
- Phasor representation of alternating voltages and currents, phasor diagrams.
- Resonance circuits analysis.
- Power relationship.

Course Materials

- **Lecture Notes and Tutorial Notes:**

Reading References:

- ***By no means can the lecture notes be replacement for the textbook:*** The student is required to carefully read and follow anyone of the following textbooks.

1. Electronics Fundamentals, Thomas Floyd, 7th ed., (Pearson, 2006) [Used by the instructor most often]
2. R.M. Mersereau, J.R. Jackson, *Circuit Analysis: A Systems Approach*, Prentice-Hall, 2006;
3. Hambley, Allan R., Electrical Engineering: Principles and Applications, Upper Saddle River, NJ: Pearson Education, Inc., 2005.

Laboratory Component:

The laboratory component has several objectives. The most important objective is to investigate experimentally the concepts taught in P238 and EE3304 course sequence. Most of the Laboratory Experiments are based on a comparison between real circuit measurements and predictions of the same circuit performance obtained by functional circuit analysis, circuit simulation using a computational model, or both.

The student is expected to confirm predictions or to observe discrepancies between predictions and measurements and draw conclusions about their causes. Therefore, as a secondary objective, the lab teaches various measurements methods and experimental skills. These include proper methods of taking data and of analyzing and presenting results, including keeping of lab logs and writing reports.
Course outcomes (think of it as what you are supposed to learn at the end of the course) are:

1. Ability to use standard laboratory equipment such as a digital multi-meter (DMM), programmable power supply (ProgPS) and arbitrary function generator (ArbFG)
2. Ability to design, build and analyze resistive and simple non linear circuits
3. Ability to design equivalent circuits (Norton/Thevenin) for maximum power
4. Ability to build 741 op-amp circuits and determine practical op-amp characteristics
5. Ability to capture, store and plot oscilloscope data

When evaluating this course, you should think whether these outcomes were actually achieved or not. Your feedback on all aspects of the labs is encouraged.

You have one week to perform each experiment in the lab. The lab report is due one week following the experiment. Prelab exercises should be worked on in the week leading up to the experiment. Although the Prelab is turned in with the lab report, it is to your advantage to complete it before the experiment is performed.

Some more details:

- Start by reading the lab assignment. This will tell you what material is covered in the experiment and which components are required to perform the experiment.
- Complete the Prelab assignment. Again, this should be done as much as possible the week before you perform the experiment in the lab.

Observe the deadlines for submitting reports!

- You are allowed to miss one deadline without any questions asked, but you need to let the lab coordinator and your TA know about this within one week of missed deadline, preferably before. You will get a ***one-week extension*** for the submission of your report. If you miss a second deadline, full documentation regarding the reasons will be required. Only unexpected emergencies will qualify. If you miss three or more you will get a failing grade.
- If you cannot come to the lab session, see your TA or send him an email note in advance. No excuses will be accepted afterwards.
- Most equipment in the lab is fixed to the bench to minimize the possibility of any damage; other items are movable and/or are checked out from the TA. Inform the TA or lab coordinator if you lack any equipment needed for the experiment. Report immediately any defective equipment or parts to the lab coordinator. As a rule, you will work in groups of two. If you have a good reason for wanting to pair up, let your TA know and he/she will decide if that can be arranged. Groups of three or more are ***not*** permitted. If more than two reports are alike, all will get zeros.

Long Syllabus

Circuits with constant voltages and currents
Convention for voltage and current; units; ohm's law, Kirchhoff's Laws and their application in series and parallel circuits. Analysis of circuits using nodal voltages; modelling of real devices with ideal elements, equivalence of voltage and current sources. Thévenin's theorem, Norton's theorem. Techniques of circuit analysis using mesh currents; power transfer from a source to a load. Linear analysis; superposition; star-delta and delta-star transformation.

Transient effects in electrical circuits
Energy storage elements; inductance and capacitance, units; types of time-varying excitation in electrical circuits, transient in circuits; modelling circuits by analytic solution of a differential equation. Exponential rise and decay, time constant in RLC circuits.

Circuits with steady-state sinusoidal excitation

Concept of frequency, angular frequency, phase shift, amplitude, peak-to-peak, and root-mean-square values. Circuit analysis suing complex mathematics; complex impedances of ideal circuit elements. Phasor representation of alternating voltages and currents, phasor diagrams.

Resonance circuit analysis; application of series and parallel resonant circuits, bandwidth and Q factor
Power relationship; Power, reactive power and VA, power factor, principle of conservation of power and reactive power, reactive power absorbed by capacitors and inductors, power factor correction, complex power in terms of phasor voltages and currents.

Basic Concepts and Laws:

Voltage, Current, Power and Energy
Circuit Elements (R, L, C, ideal operational amplifiers, ideal transformer)
Independent and Dependent Sources
Kirchhoff's Laws
Series and Parallel Combinations of Elements
Voltage Division and Current Division

Methods of Analysis

Linear Equations
Nodal Analysis
Super Nodes
Mesh Analysis
Super Meshes
Tradeoffs

Network Theorems

Linearity
Superposition
Source Transformations
Thevenin's and Norton's Theorem
Maximum Power Transfer

Sinusoidal Steady-state Analysis

Sinusoids
Phasors

Course Learning Outcomes (CLOs)

By the end of this course, students will be able to:

1. **Explain** fundamental electrical quantities (voltage, current, power, energy) and the role of circuit elements such as resistors, capacitors, inductors, sources, and operational amplifiers.
    - *Bloom's Level: Understand*
2. **Apply** Ohm's Law, Kirchhoff's Voltage and Current Laws, and other fundamental principles to analyze DC and AC circuits.
    - *Bloom's Level: Apply*
3. **Employ** systematic analysis techniques, including the Node Voltage Method and Mesh Current Method, for solving linear circuits.
    - *Bloom's Level: Apply*
4. **Analyze** the steady-state and transient behavior of circuits containing resistive, capacitive, and inductive elements.

    - *Bloom's Level: Analyze*
5. **Determine** frequency response, impedance, and transfer functions of AC circuits.
    - *Bloom's Level: Analyze*
6. **Use** mathematical tools to model, solve, and interpret the performance of linear electrical networks.
    - *Bloom's Level: Apply/Analyze*
7. **Demonstrate** problem-solving skills relevant to engineering practice through circuit analysis exercises and applications.
    - *Bloom's Level: Apply/Evaluate*
8. **Prepare** for advanced study in electronics, signal processing, power systems, and related areas of electrical engineering.
    - *Bloom's Level: Understand/Apply*